

Subject: Biology and Environmental Science (CY202)

Question Bank

Unit 4: Air and Noise Pollution

Short questions:

1. What do you mean by air pollution?
2. Define particulate matter.
3. Mention the division of particulate matter with dimension.
4. Name the major six air pollutants.
5. What are the trace pollutants?
6. What are the criteria for gaseous pollutants in urban regions?
7. What is photochemical smog? How is it different from smog?
8. What are VOCs?
9. Abbreviations: NAAQS, EPA, VOC, CFC, CCS etc.
10. What happened during the Bhopal Gas tragedy?
11. What are the optimum sizes of a PM that can be filtered by cyclones, scrubbers, and electrostatic precipitators?
12. What are baghouse filters? Mention one merit and demerit of baghouse filters.
13. What is carbon sequestration?
14. Define noise pollution.
15. What are the two basic forms of CCS?
16. What is the normal sound detection range for humans?
17. What is the range of man-made noise and environmental noise?
18. Write four harmful effects of noise pollution.
19. Determine the sound pressure level if the rms pressure value is given as $30 \mu\text{Pa}$.
20. Calculate the sound pressure level in decibels for a given sound pressure of 20×10^3 Micro-Pascal.

Broad questions:

1. Discuss the characteristics, primary source, and harmful effects of the criterion air pollutants.
2. What is acid rain? Discuss with reactions, the harmful effects of acid rain.
3. What is greenhouse effect? How do greenhouse work? Give the examples of four major greenhouse gases. Discuss the adverse effect of greenhouse effect.
4. What is global warming? Discuss the impact of greenhouse gases on global warming.

5. What is ozone layer depletion? How does CFC affect the ozone depletion?
6. Discuss the operational principle of electrostatic precipitator. Comment on its efficiency.
7. Write a short note on flue gas desulfurization.
8. Write a short note on noise pollution.